

Statistical and Machine Learning SS26

Exercise Sheet 3

Exercise 1: Confusion Matrix

A logistic regression model was trained to detect fraud in credit card transactions. The following counts were recorded on the test set:

	Predicted Positive	Predicted Negative
Actual Positive	80	20
Actual Negative	10	390

- (a) Store these values in R and compute: Accuracy, Sensitivity (TPR), Specificity, Precision, and F1-score.
 - (b) Given the context (fraud detection), which metric is most critical and why?
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Exercise 2: ROC Curve and AUC

Use the `ROCR` package (or base R) to plot a ROC curve and compute the AUC for a simulated binary classifier.

- (a) Simulate 200 observations: 100 positives and 100 negatives. For positives, draw predicted scores from $\text{Beta}(5, 2)$; for negatives, from $\text{Beta}(2, 5)$.
 - (b) Plot the ROC curve.
 - (c) Compute and interpret the AUC.
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Exercise 3: Bootstrap Standard Error

Use the bootstrap to estimate the standard error of the **median** for a sample drawn from an exponential distribution.

- (a) Draw $n = 50$ observations from $\text{Exp}(\text{rate} = 2)$.
 - (b) Compute the bootstrap standard error of the sample median using $B = 1000$ resamples.
 - (c) Compare with the analytical standard error of the mean (for reference: $SE_{\text{mean}} = \sigma/\sqrt{n}$). Comment on why bootstrap is especially useful for the median.
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Exercise 4: K-Fold Cross-Validation from Scratch

Implement 5-fold cross-validation **manually** (without using `caret` or `tidymodels`) to estimate the test MSE of a linear regression model.

- (a) Simulate data from $y = 1 + 2x - 0.5x^2 + \varepsilon$ with $n = 200$.
- (b) Fit a simple linear model ($y \sim x$) using 5-fold CV and report the average test MSE across folds.
- (c) Repeat with a quadratic model ($y \sim x + I(x^2)$). Which model has lower CV-MSE?